Math 54 — Week 6

Solutions

P1.

Idea. One row reduction gives all three bases: pivot columns of A, nonzero rows of the echelon form, and the parametric form of $A\mathbf{x} = \mathbf{0}$.

1. Row reduce:

$$\begin{bmatrix} 1 & 2 & -1 & 3 \\ 2 & 4 & 1 & 0 \\ 3 & 6 & 0 & 3 \end{bmatrix} \xrightarrow[R_3-3R_1]{R_2-2R_1} \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & 0 & 3 & -6 \\ 0 & 0 & 3 & -6 \end{bmatrix} \xrightarrow{R_3-R_2} \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & 0 & 3 & -6 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow[R_1+R_2]{\frac{1}{3}R_2} \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

2. Pivot columns 1 and 3, so a basis for $\operatorname{Col} A$ is columns 1 and 3 of A: $(1, 2, 3)$ and $(-1, 1, 0)$.
3. A basis for $\operatorname{Row} A$ is the nonzero rows of the reduced echelon form: $(1, 2, 0, 1)$ and $(0, 0, 1, -2)$.
4. x_2, x_4 are free, $x_1 = -2x_2 - x_4$ and $x_3 = 2x_4$, so

$$\mathbf{x} = x_2 \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -1 \\ 0 \\ 2 \\ 1 \end{bmatrix},$$

   and these two vectors are a basis for $\operatorname{Nul} A$.
5. Rank 2 and nullity 2: $2 + 2 = 4$, the number of columns.

P2.

Idea. Three vectors in $\mathbb{R}^3$: by the basis theorem it is enough to check independence, here by a determinant.

1. Expand along row 1 of the matrix with these columns:

$$\det \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 3 \\ -1 & 0 & 4 \end{bmatrix} = 1(4 - 0) - 2(0 + 3) + 1(0 + 1) = -1.$$

2. The determinant is not 0, so the columns are independent, and by the basis theorem they form a basis of $\mathbb{R}^3$.

P3.

Idea. Put the vectors in as columns; the pivot columns are a basis of the span (Theorem 6).

1. Row reduce:

$$\begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & -1 \\ 2 & 5 & -1 \end{bmatrix} \xrightarrow{R_3-2R_1} \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & -1 \\ 0 & 1 & -1 \end{bmatrix} \xrightarrow{R_3-R_2} \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1-2R_2} \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

2. Pivot columns 1 and 2: a basis is $\{(1, 0, 2), (2, 1, 5)\}$, and $\dim H = 2$.
3. Column 3 of the reduced form is $(2, -1, 0)$, so the third vector is 2 times the first minus the second: $2(1, 0, 2) - (2, 1, 5) = (0, -1, -1)$.

P4.

Idea. Turn $p(1) = 0$ into an equation on the coefficients, solve it, and split by free coefficient.

1. Write $p = a_0 + a_1 t + a_2 t^2 + a_3 t^3$. Then $p(1) = a_0 + a_1 + a_2 + a_3$, so the condition is $a_0 = -a_1 - a_2 - a_3$, with a_1, a_2, a_3 free.
2. Substituting and grouping by free coefficient,

$$p = (-a_1 - a_2 - a_3) + a_1 t + a_2 t^2 + a_3 t^3 = a_1(t-1) + a_2(t^2-1) + a_3(t^3-1).$$

   So $\{t-1,\ t^2-1,\ t^3-1\}$ spans the subspace.
3. The three polynomials have degrees $1, 2, 3$, so none is a combination of the ones before it (those have smaller degree); by Theorem 4 they are independent. They form a basis, and the dimension is 3.

P5.

Idea. Use the rank theorem $\operatorname{rank} A + \dim \operatorname{Nul} A = n$ and the bound $\operatorname{rank} A \le \min(m, n)$.

1. (a) $\dim \operatorname{Nul} A = 8 - 5 = 3$. $\operatorname{Col} A$ is a 5-dimensional subspace of $\mathbb{R}^5$, so by the basis theorem it is $\mathbb{R}^5$.
2. (b) No. $\operatorname{rank} A \le 4$, so $\dim \operatorname{Nul} A = 6 - \operatorname{rank} A \ge 2$.

P6.

Idea. Row reduce $[\,\mathbf{v}_1\ \mathbf{v}_2\ \mathbf{e}_1\ \mathbf{e}_2\ \mathbf{e}_3\,]$ and keep the pivot columns.

1. Row reduce:

$$\begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 2 & 1 & 0 & 1 & 0 \\ 0 & 3 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_2 - 2R_1} \begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & -2 & 1 & 0 \\ 0 & 3 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_3 - 3R_2} \begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & -2 & 1 & 0 \\ 0 & 0 & 6 & -3 & 1 \end{bmatrix}$$

2. The pivot columns are $1, 2, 3$, so a basis is $\{(1,2,0), (0,1,3), (1,0,0)\}$.

P7.

Idea. Each part is one of this week's theorems: Theorem 9, the bounds, the rank count, and the invertible matrix theorem.

1. (a) True. A basis of V has 3 vectors, and by Theorem 9 any set of more than 3 vectors in V is dependent.
2. (b) False. Two vectors may be dependent: for example $\mathbf{v}$ and $2\mathbf{v}$ span only a line, and $\mathbf{0}, \mathbf{0}$ span $\{\mathbf{0}\}$.
3. (c) True. The number of pivot columns is $\operatorname{rank} A = \dim \operatorname{Col} A$, and $\dim \operatorname{Row} A$ is the number of pivots, which is the same number.
4. (d) True. $\operatorname{Nul} A \neq \{\mathbf{0}\}$, so by statement (q) of the invertible matrix theorem A is not invertible, and then $\det A = 0$ (Week 5).